

ADDING AND SUBTRACTING NUMBERS EXPRESSED IN SCIENTIFIC NOTATION



LESSON INTRODUCTION

MA.8.NSO.1.5 Add, subtract, multiply, and divide numbers expressed in scientific notation with procedural fluency.

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

LEARNING TARGETS

- I can add and subtract numbers expressed in scientific notation.
- I can solve real-world problems involving operations with numbers expressed in scientific notation.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

WORKING TOWARD

MA.9.12.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you add and subtract numbers written in scientific notation?
- How do you express the sum or difference in scientific notation?

LESSON NARRATIVE

Students explore how to add and subtract numbers written in scientific notation. They solve mathematical and real-world problems by rewriting the numbers so the powers of 10 are the same, find the sum or difference, and then rewrite the answer in scientific notation.

COMPONENT SUMMARY

3.5.1 WARM-UP (~5 MIN.)

Notice and wonder about large numbers

3.5.3 GUIDED INSTRUCTION (15 MIN.)

Discover how to add and subtract numbers written in scientific notation

3.5.5 TRY IT (5-10 MIN.)

Additional practice adding and subtracting numbers written in scientific notation

3.5.2 EXPLORATION (5 MIN.)

Analysis of sample student work

3.5.4 GUIDED PRACTICE (10 MIN.)

Practice adding and subtracting numbers written in scientific notation

3.5.6 WRAP-UP (~5 MIN.)

Reflect on a confusing aspect of the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Scaled images of Antarctic Krill, Zooplankton, Jupiter, Saturn, and Neptune

ADDITIONAL PREPARATION

- Consider finding scaled images or three-dimensional models of the creatures in 3.5.5 Try It and planets in 3.5.2 Exploration to help students understand the context of the problems.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand why they cannot add or subtract the mantissa (numbers right of the decimal) when the powers of ten are different.
- Students may not understand how to rewrite a number so that the addends have the same power of 10.

THINGS TO CONSIDER

- Students may benefit from a review of the form of scientific notation as $a \times 10^n$ where $1 \leq |a| < 10$.
- If time is an issue, then consider assigning 3.5.5 Try It for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 3.5.1 Warm-Up prompt students to reflect on what they

notice and wonder about an image of large numbers. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and high ceiling, as students can be challenged to think of multiple noticing and wonderings.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students are provided with multiple opportunities to practice adding and subtracting numbers written in scientific notation in 3.5.3 Guided Instruction, 3.5.4 Guided Practice, and 3.5.5 Try It. This repeated practice helps students develop the knowledge and skills to solve the problems accurately and efficiently.

DIFFERENTIATE INSTRUCTION

In 3.5.2. Exploration provide students with sentence starters to help them compose their explanations. Provide an audio entry point by having students read their explanations aloud. Provide a visual and/or kinesthetic entry point by showing students scale images and/or models of the creatures in 3.5.5 Try It and planets in 3.5.2 Exploration.

Creature	Mass of One Individual (kg)
Ant	3×10^{-6}
Antarctic Krill	4.86×10^{-4}
Zooplankton	5×10^{-8}

3.5.1 WARM-UP: 25-BILLION DOWNLOADS!

TEACHER MOVES

Notice and Wonder

Prompt students to reflect on what they notice and wonder about an image of large numbers. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and high ceiling, as students can be challenged to think of multiple noticing and wonderings.

ENRICHMENT

How would you read each number? Which place values have changed? Which place values have remained the same? How do you rewrite each number in scientific notation?



3.5.1 Warm-Up: 25 Billion Downloads!

- When the Apple App Store was close to hitting 25 billion downloads, the company posted a counter on its website. As the countdown approached Daniel took the screenshots shown. Screenshot 2 was taken 4 seconds after screenshot 1.

Screenshot 1									
2	4	6	5	8	4	9	9	8	0
Screenshot 2									
2	4	6	5	8	5	0	2	2	8

- What do you notice?

Answers may vary. Possible answer – the values don't change until you reach the hundred thousands place, which made it seem like not as large of a change until you realize it's a difference of over 2,000 in just 4 seconds.

- What do you wonder?

Answers may vary. Possible answers - how is it possible for so many people to download over 2,000 apps in just 4 seconds? What apps were most popular? Globally, where were the apps being downloaded?



3.5.2 Exploration: Measuring the Planets

Work with your partner to answer the following questions.

Diego, Kiran, and Clare were discussing the question shown.

"If Neptune and Saturn were side by side, would they be wider than Jupiter?"

They determined that to answer the question they should add the diameters, 4.7×10^4 kilometers (km) and 1.2×10^5 km.

1. Review each student’s response then complete the table.

Student Response	Agree?	Explanation
Diego says, "When we add the distances, we will get $4.7 + 1.2 = 5.9$. The exponent will be 9. So, the two planets are 5.9×10^9 km side by side."	☐ Yes ☒ No	<i>They cannot be added together as they are because they have different powers of ten. Plus, we add exponents when we multiply terms, not add them.</i>
Kiran wrote 4.7×10^4 as 47,000 and 1.2×10^5 as 120,000 and added them.	☒ Yes ☐ No	<i>Each value can be written in expanded form and then added.</i>

3.5.2 EXPLORATION: MEASURING THE PLANETS

DIFFERENTIATE INSTRUCTION

Consider providing a visual and/or kinesthetic entry point by showing students scaled images and/or three-dimensional models of Neptune, Saturn, and Jupiter. Also, provide students with sentence starters to help them compose their explanations.

Clare says, "I think you can't add unless they are the same power of 10." She adds 4.7×10^4 km and 12×10^4 to get 16.7×10^4 .	☒ Yes ☐ No	<i>The tens have the same exponent, so I think you can add the 2 values then and leave the answer as $\times 10^4$.</i>
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2. Jupiter has a diameter of 1.43×10^5 . Which is wider, Neptune and Saturn put side by side, or Jupiter?
Neptune and Saturn together are wider than Jupiter, because as seen in Clare's answer, their diameter is 16.7×10^4 which can be rewritten as 1.67×10^5 .

DIFFERENTIATE INSTRUCTION

Provide an audio entry point by having students read their explanations aloud.

3.5.3 GUIDED INSTRUCTION: ADDING AND SUBTRACTING NUMBERS IN SCIENTIFIC NOTATION

TEACHER MOVES

Before beginning Guided Instruction, review with students their explanations and discussions from 3.5.2 Exploration. Kiran added by writing the equivalent standard form while Clare rewrote 1.2×10^5 as 12×10^4 so that the powers of ten are the same. Students may need guidance in grasping that 12×10^4 can be read as 12 ten thousand and 4.7×10^4 can be read as 4.7 ten thousand. Therefore, the sum is 16.7 ten thousand. Relating this to Kiran's method of writing the values using the standard form may help students understand why the power of 10 must be the same to add. Emphasizing that the place value is the reason the values of the exponents should be the same before adding. Both methods then require the answer to be revised so that it is in scientific notation. The Guided Instruction will allow for students to further grasp this.

DIFFERENTIATE INSTRUCTION

Consider prompting students to organize numbers in a place value table to provide a visual entry-point and avoid any potential confusion about the placement of the decimal or power of ten. For example, for question 2:

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3.5.3 Guided Instruction: Adding and Subtracting Numbers in Scientific Notation

1. Gizmo Gadgets produced 1,500,000 widgets in 2017. In 2018 they produced another 2,000,000.
 - a. How many widgets did they produce during the two-year period?
 $3,500,000$
 - b. Write the answer to Part A using scientific notation.
 3.5×10^6

Scientific notation can be used to add and subtract large and small numbers.

- c. Write the numbers in scientific notation.

$$1,500,000 \quad \underline{1.5} \times 10^6$$

$$2,000,000 \quad \underline{2.0} \times 10^6$$

Another way to find the sum of gadgets is to add the numbers using scientific notation. In order to add numbers using scientific notation, the power of ten should be the same.

- d. Discuss with your partner why when the power of ten are not the same that the place value is not the same. If necessary, refer back to the Exploration.
- e. Determine the sum.
 $(1.5 \times 10^6) + (2.0 \times 10^6) = 3.5 \times 10^6$

- f. Compare your answer to Part B with your answer to Part E. If they are not the same, make any necessary adjustments to your answer.

2. Write each in scientific notation.

Number	Scientific Notation
15,740,000,000	1.574×10^{10}
157.4×10^4	1.574×10^6
15.74×10^9	1.574×10^{10}
15.74×10^5	1.574×10^6

3. Which numbers in problem 2 are represented by the same scientific notation?

$15,740,000,000$ and 15.74×10^9 both represent 1.574×10^{10} , and 157.4×10^4 and 15.74×10^5 both represent 1.574×10^6 .

4. Find the sum by filling in the blanks.

$$(6.11 \times 10^5) + (1.574 \times 10^6)$$

$$(6.11 \times 10^5) + (1.574 \times 10^5)$$

$$= (21.85 \times 10^5)$$

$$= (2.185 \times 10^6)$$

5. The population of New City was 1,625,000 in 2017. The population grew to 2,750,000 by the end of 2018.
- a. How many more people lived in New City at the end of 2018 in comparison to 2017. Write your answer in scientific notation.

$$1.125 \times 10^6$$

- b. Write the population numbers from the word problem in scientific notation.

$$1,625,000 \quad 1.625 \times 10^6$$

$$2,750,000 \quad 2.75 \times 10^6$$

- c. What is the difference between the populations? Write your answer using scientific notation.

$$1.125 \times 10^6$$

6. Clare was wondering if adding and subtracting numbers in scientific notation would hold true for small numbers, as well. The measurements in the table are from her science textbook.

Item	Measurement (centimeters)
Width of a Human Hair	0.00108
Diameter of a Grain of Sand	0.00609
Length of a Dust Mite	0.0136

- a. Work with your partner to check Clare's work by writing each number in standard form, then confirm the solution.

$$(1.08 \times 10^{-3}) + (6.09 \times 10^{-3}) = (7.17 \times 10^{-3})$$

$$0.00108 + 0.00609 = 0.00717$$

- b. Find the difference between the length of a dust mite and the width of a human hair strand.

$$1.252 \times 10^{-2}$$

3.5.3 GUIDED INSTRUCTION: ADDING AND SUBTRACTING NUMBERS IN SCIENTIFIC NOTATION (CONTINUED)

TEACHER MOVES

Questions 2-4 scaffolds adding two values that are written in scientific notation by changing the place value of one value so that the power of 10 matches the other addend.

ENRICHMENT

- What final place value are we expecting in these answers?
- How does the power of 10 change as you move the decimal to the left?
- How does the power of 10 change as you move the decimal to the right?

3.5.4 GUIDED PRACTICE: ADDING AND SUBTRACTING NUMBERS WRITTEN IN SCIENTIFIC NOTATION

TEACHER MOVES

Clarification 1 of this benchmark states addition and subtraction with numbers expressed in scientific notation, exponents are limited to within two of each other. The lesson has built student understanding to within one of each other. Allow students to try question 2 before providing support to notice how well each student can transfer their learning.

DIFFERENTIATE INSTRUCTION

Prompt students to look at the model of steps outlined in question 1 and repeat the steps with the number in questions 2 and 3.

QUESTIONING

Prompt students who finish early to research real-world examples of large and small numbers and write addition and subtraction problems with the numbers written in scientific notation. Students can swap problems with a partner and solve each other's problems.



3.5.4 Guided Practice: Adding and Subtracting Numbers Written in Scientific Notation

1. Fill in the blanks to find the difference.

$$(3.8 \times 10^8) - (8.1 \times 10^7)$$

$$(\textcolor{red}{38} \times 10^7) - (8.1 \times 10^7)$$

$$\textcolor{red}{29.9} \times 10^7$$

$$\textcolor{red}{2.99} \times 10^8$$

2. Find the difference. Write your answer using scientific notation.

$$1.574 \times 10^{11} - 32.2 \times 10^9$$

$$\textcolor{red}{15.74} \times 10^{10} - \textcolor{red}{3.22} \times 10^{10} =$$

$$\textcolor{red}{12.52} \times 10^{10} =$$

$$\textcolor{red}{1.252} \times 10^{11}$$

3. Find the sum. Write your answer using scientific notation.

$$2.37 \times 10^{32} + 14.29 \times 10^{34}$$

$$\textcolor{red}{2.37} \times 10^{32} + \textcolor{red}{1429} \times 10^{32} =$$

$$\textcolor{red}{1431.37} \times 10^{32} =$$

$$\textcolor{red}{1.43137} \times 10^{35}$$

**3.5.5 Try It: Adding Or Subtracting Numbers Written in Scientific Notation.**

Add or subtract. Write your answers in scientific notation.

1. $5.3 \times 10^4 + 4.7 \times 10^4$
 $10 \times 10^4 = 1.0 \times 10^5$

2. $3.7 \times 10^6 - 3.3 \times 10^6$
 $0.4 \times 10^6 = 4.0 \times 10^5$

3. $4.8 \times 10^{-3} + 6.3 \times 10^{-3}$
 $11.1 \times 10^{-3} = 1.11 \times 10^{-2}$

4. $6.6 \times 10^{-5} - 6.1 \times 10^{-5}$
 $0.5 \times 10^{-5} = 5.0 \times 10^{-6}$

5. The table lists the masses, in kilograms (kg), of different creatures in land and in water. Use this information to answer the questions.

Creature	Mass of One Individual (kg)
Ant	3×10^{-6}
Antarctic Krill	4.86×10^{-4}
Zooplankton	5×10^{-8}

- a. What is the difference, in kg, between the mass of an ant and the mass of an Antarctic krill?
 $486 \times 10^{-6} - 3.0 \times 10^{-6}$
 $= 483.0 \times 10^{-6}$
 $= 4.83 \times 10^{-4}$
- b. What is the total mass of an ant and a zooplankton?
 $0.05 \times 10^{-6} + 3.0 \times 10^{-6}$
 $= 3.05 \times 10^{-6}$

3.5.5 TRY IT: ADDING OR SUBTRACTING NUMBERS WRITTEN IN SCIENTIFIC NOTATION**TEACHER MOVES**

Students are provided with multiple opportunities to practice adding and subtracting numbers written in scientific notation in 3.5.3 Guided Instruction, 3.5.4 Guided Practice, and 3.5.5 Try It. This repeated practice help students develop the knowledge and skills to accurately and efficiently solve the problems.

DIFFERENTIATE INSTRUCTION

Provide a visual and/or kinesthetic entry point by showing students scaled images or models of the creatures.

**3.5.6 Wrap-Up: Reflection**

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on adding and subtracting numbers expressed in scientific notation?

Answers will vary.

**3.5.6 WRAP-UP: REFLECTION****QUESTIONING**

If students were not confused during the lesson, prompt them to explain the real-world utility of the lesson learning targets.